



Choupo

The Open Source Chemical Process Simulator

Explorer Guide

Choupo-dev

Copyright © 2026 Vítor Geraldès.

Authors: Vítor Geraldès.

This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Code examples, Choupo case files, and other machine-readable examples are licensed under GPL-3.0-or-later.

Typeset in L^AT_EX.

Acknowledgment. Choupo was conceived and architected by Vítor Geraldès. Substantial parts of the initial implementation, documentation, and tutorial corpus were produced with assistance from Anthropic Claude Code using Claude Opus/Fable models. This guide is published under human curation, review, and editorial responsibility by Vítor Geraldès.

License

Attribution-ShareAlike 4.0 International

Creative Commons Corporation ("Creative Commons") is not a law firm and does not provide legal services or legal advice. Distribution of Creative Commons public licenses does not create a lawyer-client or other relationship. Creative Commons makes its licenses and related information available on an "as-is" basis. Creative Commons gives no warranties regarding its licenses, any material licensed under their terms and conditions, or any related information. Creative Commons disclaims all liability for damages resulting from their use to the fullest extent possible.

Using Creative Commons Public Licenses

Creative Commons public licenses provide a standard set of terms and conditions that creators and other rights holders may use to share original works of authorship and other material subject to copyright and certain other rights specified in the public license below. The following considerations are for informational purposes only, are not exhaustive, and do not form part of our licenses.

Considerations for licensors: Our public licenses are intended for use by those authorized to give the public permission to use material in ways otherwise restricted by copyright and certain other rights. Our licenses are irrevocable. Licensors should read and understand the terms and conditions of the license they choose before applying it. Licensors should also secure all rights necessary before applying our licenses so that the public can reuse the material as expected. Licensors should clearly mark any material not subject to the license. This includes other CC-licensed material, or material used under an exception or limitation to copyright. More considerations for licensors: wiki.creativecommons.org/Considerations_for_licensors

Considerations for the public: By using one of our public licenses, a licensor grants the public permission to use the licensed material under specified terms and conditions. If the licensor's permission is not necessary for any reason--for example, because of any applicable exception or limitation to copyright--then that use is not regulated by the license. Our licenses grant only permissions under copyright and certain other rights that a licensor has authority to grant. Use of the licensed material may still be restricted for other reasons, including because others have copyright or other rights in the material. A licensor may make special requests, such as asking that all changes be marked or described. Although not required by our licenses, you are encouraged to respect those requests where reasonable. More considerations for the public: wiki.creativecommons.org/Considerations_for_licensees

Creative Commons Attribution-ShareAlike 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree to be bound by the terms and conditions of this Creative Commons Attribution-ShareAlike 4.0 International Public License ("Public License"). To the extent this Public License may be interpreted as a contract, You are granted the Licensed Rights in consideration of Your acceptance of these terms and conditions, and the Licensor grants You such rights in consideration of benefits the Licensor receives from making the Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

- a. Adapted Material means material subject to Copyright and Similar Rights that is derived from or based upon the Licensed Material and in which the Licensed Material is translated, altered, arranged, transformed, or otherwise modified in a manner requiring permission under the Copyright and Similar Rights held by the Licensor. For purposes of this Public License, where the Licensed Material is a musical work, performance, or sound recording, Adapted Material is always produced where the Licensed Material is synched in timed relation with a moving image.
- b. Adapter's License means the license You apply to Your Copyright and Similar Rights in Your contributions to Adapted Material in accordance with the terms and conditions of this Public License.
- c. BY-SA Compatible License means a license listed at creativecommons.org/compatiblelicenses, approved by Creative Commons as essentially the equivalent of this Public License.
- d. Copyright and Similar Rights means copyright and/or similar rights closely related to copyright including, without limitation, performance, broadcast, sound recording, and Sui Generis Database Rights, without regard to how the rights are labeled or categorized. For purposes of this Public License, the rights specified in Section 2(b)(1)-(2) are not Copyright and Similar

Rights.

- e. Effective Technological Measures means those measures that, in the absence of proper authority, may not be circumvented under laws fulfilling obligations under Article 11 of the WIPO Copyright Treaty adopted on December 20, 1996, and/or similar international agreements.
- f. Exceptions and Limitations means fair use, fair dealing, and/or any other exception or limitation to Copyright and Similar Rights that applies to Your use of the Licensed Material.
- g. License Elements means the license attributes listed in the name of a Creative Commons Public License. The License Elements of this Public License are Attribution and ShareAlike.
- h. Licensed Material means the artistic or literary work, database, or other material to which the Licensor applied this Public License.
- i. Licensed Rights means the rights granted to You subject to the terms and conditions of this Public License, which are limited to all Copyright and Similar Rights that apply to Your use of the Licensed Material and that the Licensor has authority to license.
- j. Licensor means the individual(s) or entity(ies) granting rights under this Public License.
- k. Share means to provide material to the public by any means or process that requires permission under the Licensed Rights, such as reproduction, public display, public performance, distribution, dissemination, communication, or importation, and to make material available to the public including in ways that members of the public may access the material from a place and at a time individually chosen by them.
- l. Sui Generis Database Rights means rights other than copyright resulting from Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, as amended and/or succeeded, as well as other essentially equivalent rights anywhere in the world.
- m. You means the individual or entity exercising the Licensed Rights under this Public License. Your has a corresponding meaning.

Section 2 -- Scope.

- a. License grant.
 - 1. Subject to the terms and conditions of this Public License, the Licensor hereby grants You a worldwide, royalty-free, non-sublicensable, non-exclusive, irrevocable license to exercise the Licensed Rights in the Licensed Material to:
 - a. reproduce and Share the Licensed Material, in whole or in part; and
 - b. produce, reproduce, and Share Adapted Material.
 - 2. Exceptions and Limitations. For the avoidance of doubt, where Exceptions and Limitations apply to Your use, this Public License does not apply, and You do not need to comply with its terms and conditions.
 - 3. Term. The term of this Public License is specified in Section 6(a).
 - 4. Media and formats; technical modifications allowed. The Licensor authorizes You to exercise the Licensed Rights in all media and formats whether now known or hereafter created, and to make technical modifications necessary to do so. The Licensor waives and/or agrees not to assert any right or authority to forbid You from making technical modifications necessary to exercise the Licensed Rights, including technical modifications necessary to circumvent Effective Technological Measures. For purposes of this Public License, simply making modifications authorized by this Section 2(a) (4) never produces Adapted Material.
 - 5. Downstream recipients.
 - a. Offer from the Licensor -- Licensed Material. Every recipient of the Licensed Material automatically receives an offer from the Licensor to exercise the Licensed Rights under the terms and conditions of this Public License.
 - b. Additional offer from the Licensor -- Adapted Material. Every recipient of Adapted Material from You automatically receives an offer from the Licensor to exercise the Licensed Rights in the Adapted Material under the conditions of the Adapter's License You apply.
 - c. No downstream restrictions. You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, the Licensed Material if doing so restricts exercise of the Licensed Rights by any recipient of the Licensed Material.

6. No endorsement. Nothing in this Public License constitutes or may be construed as permission to assert or imply that You are, or that Your use of the Licensed Material is, connected with, or sponsored, endorsed, or granted official status by, the Licensor or others designated to receive attribution as provided in Section 3(a)(1)(A)(i).

b. Other rights.

1. Moral rights, such as the right of integrity, are not licensed under this Public License, nor are publicity, privacy, and/or other similar personality rights; however, to the extent possible, the Licensor waives and/or agrees not to assert any such rights held by the Licensor to the limited extent necessary to allow You to exercise the Licensed Rights, but not otherwise.
2. Patent and trademark rights are not licensed under this Public License.
3. To the extent possible, the Licensor waives any right to collect royalties from You for the exercise of the Licensed Rights, whether directly or through a collecting society under any voluntary or waivable statutory or compulsory licensing scheme. In all other cases the Licensor expressly reserves any right to collect such royalties.

Section 3 -- License Conditions.

Your exercise of the Licensed Rights is expressly made subject to the following conditions.

a. Attribution.

1. If You Share the Licensed Material (including in modified form), You must:
 - a. retain the following if it is supplied by the Licensor with the Licensed Material:
 - i. identification of the creator(s) of the Licensed Material and any others designated to receive attribution, in any reasonable manner requested by the Licensor (including by pseudonym if designated);
 - ii. a copyright notice;
 - iii. a notice that refers to this Public License;
 - iv. a notice that refers to the disclaimer of warranties;
 - v. a URI or hyperlink to the Licensed Material to the extent reasonably practicable;
 - b. indicate if You modified the Licensed Material and retain an indication of any previous modifications; and
 - c. indicate the Licensed Material is licensed under this Public License, and include the text of, or the URI or hyperlink to, this Public License.
2. You may satisfy the conditions in Section 3(a)(1) in any reasonable manner based on the medium, means, and context in which You Share the Licensed Material. For example, it may be reasonable to satisfy the conditions by providing a URI or hyperlink to a resource that includes the required information.
3. If requested by the Licensor, You must remove any of the information required by Section 3(a)(1)(A) to the extent reasonably practicable.

b. ShareAlike.

In addition to the conditions in Section 3(a), if You Share Adapted Material You produce, the following conditions also apply.

1. The Adapter's License You apply must be a Creative Commons license with the same License Elements, this version or later, or a BY-SA Compatible License.
2. You must include the text of, or the URI or hyperlink to, the Adapter's License You apply. You may satisfy this condition in any reasonable manner based on the medium, means, and context in which You Share Adapted Material.
3. You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, Adapted Material that restrict exercise of the rights granted under the Adapter's License You apply.

Section 4 -- Sui Generis Database Rights.

Where the Licensed Rights include Sui Generis Database Rights that apply to Your use of the Licensed Material:

- a. for the avoidance of doubt, Section 2(a)(1) grants You the right to extract, reuse, reproduce, and Share all or a substantial portion of the contents of the database;
- b. if You include all or a substantial portion of the database contents in a database in which You have Sui Generis Database Rights, then the database in which You have Sui Generis Database Rights (but not its individual contents) is Adapted Material, including for purposes of Section 3(b); and
- c. You must comply with the conditions in Section 3(a) if You Share all or a substantial portion of the contents of the database.

For the avoidance of doubt, this Section 4 supplements and does not replace Your obligations under this Public License where the Licensed Rights include other Copyright and Similar Rights.

Section 5 -- Disclaimer of Warranties and Limitation of Liability.

- a. UNLESS OTHERWISE SEPARATELY UNDERTAKEN BY THE LICENSOR, TO THE EXTENT POSSIBLE, THE LICENSOR OFFERS THE LICENSED MATERIAL AS-IS AND AS-AVAILABLE, AND MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND CONCERNING THE LICENSED MATERIAL, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHER. THIS INCLUDES, WITHOUT LIMITATION, WARRANTIES OF TITLE, MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, ABSENCE OF LATENT OR OTHER DEFECTS, ACCURACY, OR THE PRESENCE OR ABSENCE OF ERRORS, WHETHER OR NOT KNOWN OR DISCOVERABLE. WHERE DISCLAIMERS OF WARRANTIES ARE NOT ALLOWED IN FULL OR IN PART, THIS DISCLAIMER MAY NOT APPLY TO YOU.
- b. TO THE EXTENT POSSIBLE, IN NO EVENT WILL THE LICENSOR BE LIABLE TO YOU ON ANY LEGAL THEORY (INCLUDING, WITHOUT LIMITATION, NEGLIGENCE) OR OTHERWISE FOR ANY DIRECT, SPECIAL, INDIRECT, INCIDENTAL, CONSEQUENTIAL, PUNITIVE, EXEMPLARY, OR OTHER LOSSES, COSTS, EXPENSES, OR DAMAGES ARISING OUT OF THIS PUBLIC LICENSE OR USE OF THE LICENSED MATERIAL, EVEN IF THE LICENSOR HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH LOSSES, COSTS, EXPENSES, OR DAMAGES. WHERE A LIMITATION OF LIABILITY IS NOT ALLOWED IN FULL OR IN PART, THIS LIMITATION MAY NOT APPLY TO YOU.
- c. The disclaimer of warranties and limitation of liability provided above shall be interpreted in a manner that, to the extent possible, most closely approximates an absolute disclaimer and waiver of all liability.

Section 6 -- Term and Termination.

- a. This Public License applies for the term of the Copyright and Similar Rights licensed here. However, if You fail to comply with this Public License, then Your rights under this Public License terminate automatically.
- b. Where Your right to use the Licensed Material has terminated under Section 6(a), it reinstates:
 1. automatically as of the date the violation is cured, provided it is cured within 30 days of Your discovery of the violation; or
 2. upon express reinstatement by the Licensor.

For the avoidance of doubt, this Section 6(b) does not affect any right the Licensor may have to seek remedies for Your violations of this Public License.

- c. For the avoidance of doubt, the Licensor may also offer the Licensed Material under separate terms or conditions or stop distributing the Licensed Material at any time; however, doing so will not terminate this Public License.
- d. Sections 1, 5, 6, 7, and 8 survive termination of this Public License.

Section 7 -- Other Terms and Conditions.

- a. The Licensor shall not be bound by any additional or different terms or conditions communicated by You unless expressly agreed.
- b. Any arrangements, understandings, or agreements regarding the Licensed Material not stated herein are separate from and independent of the terms and conditions of this Public License.

Section 8 -- Interpretation.

- a. For the avoidance of doubt, this Public License does not, and shall not be interpreted to, reduce, limit, restrict, or impose conditions on any use of the Licensed Material that could lawfully be made without permission under this Public License.
- b. To the extent possible, if any provision of this Public License is deemed unenforceable, it shall be automatically reformed to the minimum extent necessary to make it enforceable. If the provision cannot be reformed, it shall be severed from this Public License without affecting the enforceability of the remaining terms and conditions.

- c. No term or condition of this Public License will be waived and no failure to comply consented to unless expressly agreed to by the Licensor.
- d. Nothing in this Public License constitutes or may be interpreted as a limitation upon, or waiver of, any privileges and immunities that apply to the Licensor or You, including from the legal processes of any jurisdiction or authority.

=====
Creative Commons is not a party to its public licenses. Notwithstanding, Creative Commons may elect to apply one of its public licenses to material it publishes and in those instances will be considered the "Licensor." The text of the Creative Commons public licenses is dedicated to the public domain under the CC0 Public Domain Dedication. Except for the limited purpose of indicating that material is shared under a Creative Commons public license or as otherwise permitted by the Creative Commons policies published at creativecommons.org/policies, Creative Commons does not authorize the use of the trademark "Creative Commons" or any other trademark or logo of Creative Commons without its prior written consent including, without limitation, in connection with any unauthorized modifications to any of its public licenses or any other arrangements, understandings, or agreements concerning use of licensed material. For the avoidance of doubt, this paragraph does not form part of the public licenses.

Creative Commons may be contacted at creativecommons.org.

Trademarks

Choupo is a trademark of TalentGround Lda.

Anthropic, Claude, and Claude Code are trademarks or registered trademarks of Anthropic, PBC.

1 What the Explorer is

The Explorer is Choupo’s *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: “**what does this thermodynamics look like?**”

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway γ) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no “display model”: if the Explorer shows an azeotrope at $x = 0.89$, the distillation column will hit it at $x = 0.89$.

2 The workflow

1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals, P^{sat} , c_p , formation — and what it lacks). Case-local components (from an open case's **constant**/) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state, P^{sat} correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

3 The views, one by one

3.1 The compound browser (the Explorer's landing)

The catalogue is a *tree*, not a list. Groups derived from each record's own facts fold and unfold with their member count; the groups a student charges to a vessel open by default, the combustion library, the radicals and the synthetic test stand-ins start folded. A group too large for one screen is subdivided into *families*, and the family comes from the record, never from the name: first the *declared UNIFAC groups* (the highest-precedence group present decides — an acid before an ester before an alcohol, down to an alkane); only for a record with no group decomposition, the *elemental formula*, coarser and labelled so with a “(by formula)” suffix. A record with neither is shown under “*No formula declared in the record*” — a visible curation gap, never a guess. Typing in the search box flattens the tree; a double click inspects a component without changing the set. The browser reads the catalogue; it does not edit a record.

The browser is the left half of the Explorer's *landing* — the screen `?workspace=explore` opens; its right half shows the record of whichever compound you last touched — identity, what the record carries, the curation dossier's verdict — and its **Explore properties** button opens the property surfaces in a tab of their own, carrying the set in the address (`?workspace=properties&components=water,ethanol`), which makes an exploration a link you can hand to somebody. Neither tab needs the other: the surfaces have their own door on the hub, keep a compound rail, and open perfectly well with nothing selected, and a component the address names but the catalogue cannot resolve is left out of the set and said so, never dropped in silence.

3.2 Property vs T/P (scan)

Any pure-component property — P^{sat} , liquid density, c_p , viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

3.3 Pure phase diagram ($P-T$)

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

3.4 Binary boiling envelope ($T-x-y$)

The bubble and dew curves of a binary at fixed P . This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal $\rightarrow$ NRTL $\rightarrow$ UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

3.5 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary: γ_1, γ_2 against composition, with the infinite-dilution values γ^∞ at the edges. $\gamma > 1$: the pair dislikes mixing (positive deviation, minimum-boiling azeotropes); $\gamma < 1$: association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

3.6 Binary flash ($x-y$ + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy: $g_{\text{mix}}(x)$ with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

3.8 Ternary boiling surface

T_{bubble} over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

3.9 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward $\text{SI} = 0$ (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

3.10 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the $T \times P$ plane. The whole point is the tension it makes visible: for ammonia ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first (“where do I write the reaction?” — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach ΔT (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts the $\Delta T = 0$ contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

3.11 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 — h_f, h_g, h_{fg} , entropies, volumes — the backbone of every steam and power-cycle case, browsable.

3.12 EduTools (the Methods workspace)

Since 2026-08-15 the classical *method constructions* live in their own workspace, **EduTools**, split from the Explorer by one criterion: *a property surface answers “what does this system's property look like?”* (the Explorer); *a method construction answers “what does the classical graphical method say about a design question?”* — McCabe–Thiele, Kremser, Rayleigh, ε -NTU, pinch composites, and derivation pages for the property models. Open it from the hub's **EduTools** button: it is a tab of its own, needs no case, and does not talk to the flowsheet. A tool is chosen from the top row of the tab; every tool carries a one-line “teaches” caption and its Theory Guide link. Every curve is an engine run (`choupoProps` in `WebAssembly`) over a transient synthesized case — *zero physics in TypeScript*; the staircase drawn on it is geometry. Planned tools stay visible and disabled, naming what will feed them. It never writes a case: “Author → copy propsDict” emits the dict for you to author on disk.

3.13 Literature (“who measured this?”)

Provenance is the differentiator, and this workspace puts its question one click from the flowsheet. With a case open, the tab's menu carries **Literature**: type compound names (every word must match a compound in the article) or click a chip for one of the case's own components, and read who measured that system — authors, year, title, journal, a DOI link, and the properties measured. It searches your *private* mirror of the NIST/TRC ThermoML Archive, installed once with `bin/choupo-thermoml sync` on your own machine; the runtime never reads it and the public site has no mirror, so there it shows the install instructions instead. That absence is a status, not an empty result: “*nobody measured this*” and “*you have no mirror*” are different sentences. It is a *reading list*, never a *value*: no number is copied into a case or a record, and the only outbound action is opening the article. Read it and cite it — never this index.

4 Provenance and honesty rules

The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

5 From Explorer to case: the hand-off

A typical session: you notice in the T - x - y view that your solvent pair azeotropes at 78% — so the simple column in your flowsheet cannot deliver 99% purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.